

CS15-2 COMPARISON OF DIFFERENT TYPES OF TRUE AND FAKE NEWS, BOTH IN TEXT AND VISUALS

Project Proposal



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Group Based 5703 Capstone Project

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ABSTRACT

This project focuses on research papers that have been withdrawn, and compares them with those that have not been withdrawn in terms of text and vision to study the difference between real information and false information. The motivation of the project comes from the increasing number of unreliable or misleading scientific content, so it is necessary to understand the different characteristics of such content more deeply, especially in the context of academic communication and network communication. This study aims to find a model that can distinguish withdrawn papers from reliable research results by treating them as an agent of "false" or problematic information.

This project will build a large-scale dataset based on about 65000 papers that have been withdrawn, and combine structured information collected from multiple sources, including DOI pages, PubMed (PMID) and Altmetric platforms. The archived MHTML web page will be used to retain the original presentation form of the paper, so that it can not only analyze text content such as abstract and body, but also analyze visual and structural elements such as layout, image and metadata presentation.

The main goal of this project is to identify and compare the key features related to real scientific information and false scientific information, and explore the differences between the withdrawn papers and reliable papers in terms of language use, content structure and visual presentation. To achieve this goal, this project will combine natural language processing technology and multimodal analysis methods to conduct comprehensive research on text signals and visual signals.

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1. INTRODUCTION

There is a surge in the volume of fake news in the current digital society, and it has become an increasingly important issue. With the rapid growth of social media platforms, online news websites and even AI chatbots, people are exposed to a huge amount of news and information every day, but not all of it is trustworthy. Furthermore, false or misleading news can spread very quickly, influence public opinion, create confusion and might even cause harmful actions. For instance, many people use LLMs like ChatGPT as a search engine. However, ChatGPT cannot even identify retracted articles(Thelwall et al., 2025). Consequently, it may generate false statements or content for users, which could cause them to form incorrect perceptions of the things they searched for. This makes it important to better understand how fake news differs from true news and how these differences can be recognised.

This project would focus on the fake news that used or quoted the retracted articles, and compare them with the true news from both textual and visual perspectives. The project aims to discover the nature of true and fake news and identify the main textual and visual characteristics that distinguish true news from fake news. The motivation comes from the growing challenges caused by misleading information in everyday life. The news used an academic paper, but later the paper was retracted. The news might still be there and presented in a convincing way, it can even be difficult for ordinary readers to judge whether the information is reliable. In addition, the problem becomes more serious when misleading images are used along with the news, as the combination can make false information appear more believable. Another motivation for the project is that fake news does not always look the same across different topics (politics, health, entertainment). This project would highlight category-specific patterns, which can provide a broader understanding of how false information is created in different contexts.

By addressing this problem, people can improve their general understanding of how true and fake news differ in practice and the general patterns of text and images in fake news. Moreover, it might support the development of better tools for identifying misleading or fake information. Overall, this project would help people receive information more rationally in the modern digital society to reduce the negative impact of fake news.

2. RELATED LITERATURE

This proposal focuses on the use or citation of misleading news of withdrawn articles, and compares them with real news through text and visual evidence. Existing studies show that false information is rarely generated by a single signal, which usually depends on the combined effect of language, presentation, information source credibility and cross-modal consistency. Therefore, it is necessary to conduct a literature review to identify the main text and visual features that distinguish between credible information and misleading content, clarify the most commonly used methods and data sets, and explain why multimodal methods are needed. This review examines the research on the analysis of false news, multimodal false information, research on withdrawn manuscripts and media dissemination after manuscript withdrawal. The main research question is: to what extent does the existing literature explain the difference in text, visual and credibility between credible information and misleading news that relies on withdrawn manuscripts?

2.1 Literature Review

The selected research was mainly published between 2018 and 2025, during which the research on calculating false news developed rapidly. Broadly speaking, the study of fake news has gone beyond simple text classification. At present, this field usually revolves around false knowledge, writing style, communication mode and source credibility (Zhou & Zafarani, 2020). This broader perspective is valuable because it treats fake news as a credibility issue, rather than merely a NLP task. At the same time, the quality of the data set has a significant impact on any conclusion drawn from the detection model. Dataset labels, balance, domain and authenticity have a significant impact on evaluation results (D'Ulizia et al., 2021). FakeNewsNet addresses this issue by linking news content to social context and temporal information, resulting in comparisons that are more realistic than those based on plain text alone (Shu et al., 2020). In short, reliable comparison depends not just on the model, but also on whether the data really matches the real information environment.

Literature on text differences shows that false or misleading news often has a recognisable style. There are differences in language patterns in credible news, propaganda, pranks and satirical works, which means that deceptive content can

usually be distinguished by language use and rhetorical structure (Rashkin et al., 2017). Style-based detection research reinforces this argument, but also raises an important warning: the model may learn information about the publisher's identity or specific events, rather than the more common deceptive characteristics (Przybyła, 2020). Other studies have identified clues related to credibility, such as punctuation marks, capitalisation, pronoun use and emotional characteristics (Hardalov et al., 2016). Together, these studies show that misleading content is usually more sensational, emotional and more unstable rhetoric than legitimate reports. However, these studies also reveal a limitation. Text signals usually depend on the subject and may be difficult to migrate across fields. This restriction is particularly important here, because the news that quotes the withdrawn papers may still be fluent and scientifically credible at the language level.

Visual and multimodal research solves this problem more directly. Image reuse can make false statements more convincing, even if the image itself is real, because the deception lies in the pairing of the image and the statement, not just the pixel itself (Sabir et al., 2018). The study of image verification out of context further expands this view. The study uses external online evidence to check whether the image-descriptive pair is misused (Abdelnabi et al., 2022). Event-invariant multimodal learning also improves the generalisation ability across themes, indicating that text-image comparison can reduce over-fitting of specific events (Wang et al., 2018). More recent reviews point out that false information is increasingly tending to be multimodal, and the key challenge often lies in the relationship between text and images, rather than single-modal (Abdali et al., 2024). At the same time, the multimodal benchmark itself is not always reliable. Some systems have single-modal deviations, so that they can be effective without really reasoning on text and images (Papadopoulos et al., 2024). These studies have drawn a clear conclusion: visual misleading information is often persuasive because real images are placed in a false context; and multimodal analysis is not only crucial to prediction, but also important for interpretation.

Early studies have shown that the withdrawn research may still be cited (Pfeifer & Snodgrass, 1990). Updated research shows that this problem has now spread to digital systems. Withdrawn papers will still attract the attention of the media and social media, and the original articles are often more eye-catching than the

withdrawal statement (Serghiou et al., 2021). Hsiao and Schneider (2021) further pointed out that many withdrawals do not mention withdrawals at all, which helps explain why corrected scientific research still contributes to misleading narratives. This concern now extends to digital tools. ChatGPT may not mention withdrawal and other reliability issues when evaluating problem-based assertions (Thelwall et al., 2025). News research also points in the same direction, indicating that scientific news is not always updated in time after the withdrawal of manuscripts (Valinciute & Halfman, 2025). This study adds an important dimension to the broader field of fake news. Misinformation may come not only from fabricated news, but also from the continuous reuse of proven scientific assertions in seemingly legitimate information channels.

In general, the existing research has shown four main aspects. First of all, the difference between misleading news and reliable news often lies in the combined role of a variety of factors such as style, structure and credibility, rather than a single isolated feature. Secondly, visual misleading information usually relies on context mismatch, image reuse and multimodal inconsistency, not just obvious editing. Third, compared with pure text or pure image methods, multimodal methods can provide a more realistic analysis basis. Fourth, even after formal correction, the withdrawn research may continue to be disseminated in the news and digital system. However, few studies clearly explain how these signals interact in misleading news. This research gap is the necessity of this study, and also shows that future research needs to conduct more explainable multimodal comparisons.

3. PROJECT PROBLEMS DRAFTING BASED ON INITIAL DETAILS FROM CLIENT

During the age of globalization and internet booming, the spread of online information has made it kindly difficult for readers to identify the reliability of the contents. This challenge becomes more serious when news articles cite academic studies, because scientific papers can lend a strong appearance of credibility even when the underlying research has later been retracted or questioned. The current proposal already identifies this risk, noting that fake or misleading news may continue circulating even after the source paper has been retracted, and that misleading visuals can further increase perceived credibility.

The project can be seen as a comparison of true and fake news in both textual and visual forms. According to the project brief, they expected us to conduct dataset development, textual analysis, image analysis, category-based comparison, and an integrated framework for understanding differences between real and fake news. However, without first establishing a reliable and traceable dataset, we cannot directly address this overall problem within one capstone semester.

A further complication is that two student groups, CS15-1 and CS15-2, are working on the same overall project, and the client has indicated that each group will eventually be assigned a different analytical direction. In the client's latest email, the temporary decision is that the first three parts of the work are common across both groups: cleaning the preliminary CSV dataset, using the four identifier columns to collect and preprocess MHTML files, and collecting Altmetric and/or PlumX details pages. Only the later analysis stage is expected to diverge into two possible directions.

Importantly, this temporary direction was communicated only on the proposal submission date, while the proposal milestone in the current schedule is due in Week 5 on 29-03-2026. As a result, the present proposal cannot responsibly commit to one final group-specific analytical direction. Instead, it must define the project problem around the common technical and data-preparation tasks that have already been confirmed by the client. This timing issue is also relevant to the justification of scope, because later-stage analysis, methods, and deliverables may still need revision after the client formally allocates the two groups to different directions.

Therefore, the immediate project problem is defined as follows:

How can a reliable and well-linked base dataset be built from the client-provided retraction-related records, so that it can support later analysis of true and fake news in whichever group-specific direction is assigned?

This problem is practically necessary for the current stage of the project. The dataset provided by client contains structured metadata for retracted records, including fields such as RetractionDOI, RetractionPubMedID, OriginalPaperDOI, and OriginalPaperPubMedID, which can be used to build publisher and PubMed links, get and archive pages in MHTML format, and retrieve Altmetric or related attention data. The current proposal already reflects this orientation in its project

problems, methodology, and milestone plan, especially the emphasis on creating a base dataset before deeper textual or visual analysis begins.

The importance of retracted research as a motivating context is supported by the literature. Thelwall et al. (2025) found that ChatGPT did not reference retractions in 6,510 research-quality outputs. For 61 claims from retracted articles, it often judged them to be true, partially true, or in line with research. This shows that problematic academic content may still affect users through digital systems, which highlights the need for a carefully curated and trustworthy dataset before doing any downstream comparison or model-based analysis.

3.1 Project Aims & Objectives

The main aim of this project is to establish a robust and traceable data foundation for the broader study of true and fake news in textual and visual forms, while accommodating the fact that the final analytical direction for this group is still pending client confirmation. This is consistent with the current proposal, which already prioritises data retrieval and preprocessing in the earlier milestones before moving to analysis..

At the current stage, the project objectives are:

- To clean and prepare the preliminary client-provided CSV dataset.
- To identify and validate the four key identifier fields used for retrieval and linking.
- To construct a reproducible pipeline for collecting and preprocessing MHTML webpage snapshots from DOI- and PMID-related sources.
- To retrieve Altmetric and/or PlumX details-page data and link all collected outputs back to the same original records.
- To organise the returned data in a machine-readable and traceable structure for later use.
- To prepare a high-quality shared base dataset that supports both groups' common work before the final analytical split.
- To preserve flexibility so that the later analysis can be adapted once the client formally confirms the direction assigned to this group.

Completion of this stage will be concluded by whether the group can produce a clean and linked dataset, store MHTML and Altmetric-related outputs in a consistent

structure, and demonstrate that the resulting data are suitable for whichever direction is formally allocated after the client meeting.

3.2 Project Questions

According to the latest client meeting, the project questions can be summed up below:

- How can the preliminary client dataset be cleaned and standardized so that the relevant identifier fields can be used reliably?
- What is the most appropriate pipeline for collecting and preprocessing MHTML snapshots using the DOI and PubMed-based identifiers provided in the dataset?
- How can Altmetric and/or PlumX detail-page data be retrieved, structured, and linked back to the same records?
- What issues of missing identifiers, inaccessible pages, storage overhead, or incomplete attention data are likely to affect the quality of the resulting dataset?
- How can the common base dataset be designed so that it supports both possible later directions, namely multi-LLM analysis of retracted research or Altmetrics-based comparison of true and fake news?

These questions are more defensible than prematurely committing to only one final analysis path. They also better reflect the actual project state at the time of proposal submission, given that the client's temporary direction was only communicated on the submission date and formal direction allocation is still pending.

3.3 Project Scope

The scope of this project is intentionally defined around the shared foundational work confirmed by the client at the time of proposal submission.

Included in scope at this stage are: examination and cleaning of the preliminary dataset; validation of the four main retrieval identifiers; collection and preprocessing of MHTML webpage snapshots; retrieval of Altmetric and/or PlumX details-page data; organisation and linkage of all outputs to the original records; and preparation of a reusable base dataset for later analytical work.

Provisionally connected but not yet fully fixed in scope are the later-stage analytical tasks. According to the client's temporary decision, the analysis stage may follow one of two directions: one direction is similar to Thelwall et al. (2025), extending the evaluation to multiple LLMs; the other direction uses Altmetrics data about news and blogs to conduct further analysis of true and fake news based on images, headlines, and text. Because the client has not yet formally assigned which group will take which direction, the exact later analysis design cannot be finalised in this proposal.

Outside the current confirmed scope are: a complete production-ready misinformation detection system; final conclusions about textual and visual differences before the direction allocation meeting; and any fully specified downstream methodology that depends on the yet-to-be-confirmed group assignment. This limitation is not due to lack of planning, but to the late timing of the temporary client decision relative to the proposal deadline.

4. METHODOLOGIES

4.1 Methods

This project uses a data-driven, multimodal analytical methodology to investigate the differences between true and fake news. The overall approach aligns with a structured pipeline including data collection, data cleaning and organisation, and multimodal analysis using large language models.

The process begins with large-scale data collection. The provided original dataset are scientific papers retrieved from multiple sources, including publisher webpages via DOI, PubMed and records via PMID. For each paper, the corresponding MHTML webpages and external news data are collected to preserve both the textual content and visual structure, which will form a comprehensive multimodal dataset. After data collection, a data cleaning and organization phase is conducted to ensure dataset quality and consistency. This includes validating the completeness of each record, identifying and removing missing, invalid, or erroneous webpages. Then ensuring that each paper contains the required set of MHTML files and associated Altmetric detail page data. Only papers with complete and valid

multimodal information are retained for future analysis, thereby reducing noise and improving the reliability of the dataset.

After preprocessing, the project applies multimodal large language model techniques to analyse and assess scientific content. By integrating textual, visual, and structural features extracted from MHTML documents, together with external signals such as news coverage. The multimodal LLM will be used to evaluate the credibility of scientific papers. In this framework, retracted papers are treated as instances of fake scientific information. The model is employed to analyse whether such unreliable content can be effectively identified. This approach enables a systematic investigation of how multimodal information contributes to distinguishing between reliable and unreliable research. As well as the limitations of current LLM-based systems in handling scientifically questionable content.

4.2 Data Collection

This project adopts the strategy of multi-source heterogeneous data fusion for data collection. Aiming to build a comprehensive analysis library that contains both original academic metadata and network influence indicators. The core data is derived from the provided data set, which we will extract the key unique identifier of each retraction record, mainly including the retraction DOI, the original paper DOI, and the corresponding PubMedID.

Based on these identifiers, the data collection process is divided into two parallel stages: web snapshot archiving and social attention quantification. In the first phase, using the Playwright asynchronous browser automation framework, the system will construct a standard query URL of the publisher or PubMed based on the identifier, automatically access and render the page, and automatically render the page, finally, the complete dynamic web page is saved as a snapshot in MHTML format. The MHTML format was chosen because it preserves the CSS and JavaScript rendering of the page and ensures the readability of the archived content and the validity of the evidence. In the second phase, by calling the Altmetric Explorer API, the system retrieves and extracts detailed social impact metadata, indexed by the title or identifier of the paper, specific indicators include Altmetric composite scores, news

media mentions, blog citations, social media like X and Facebook, policy document citations, and Wikipedia citations.

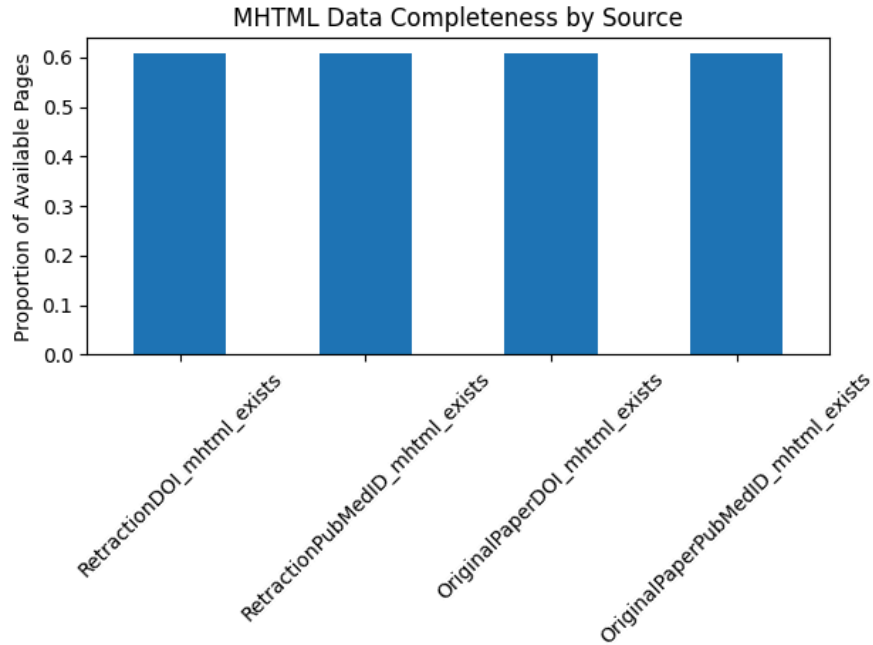
In order to solve the problem of missing identifiers of some records in the dataset, the code has built-in cleaning logic, which can automatically identify null values or invalid formats. Then preferentially match the original paper identifier to ensure data relevance. Finally, all the structured metrics are summarized and stored as CSV files, while the unstructured snapshots are stored hierarchically according to the record ID to form a traceable data link.

4.3 Data Analysis

The data analysis process in this project begins with a validation of the outputs generated from the crawling pipeline. Specifically, the MHTML records collected during crawling are examined to identify any missing, invalid, or erroneous webpages. This step is essential to ensure the integrity and consistency of the dataset before any further analysis. For each retracted paper, the project requires a complete set of records associated with four identifiers: Retraction DOI, Retraction PubMed ID, Original Paper DOI, and Original Paper PubMed ID, together with valid Altmetric details page data. These five components form the minimum data requirement for inclusion in the dataset. During analysis, each crawled record is checked to confirm that all four MHTML pages have been successfully retrieved and the corresponding Altmetric details pages are valid and usable. Pages that are inaccessible, corrupted, empty, or incorrectly matched are treated as invalid records.

To ensure dataset quality, papers with any missing component in these five required contents are removed from the final analytical dataset. In other words, only retracted papers with a complete and valid set of four MHTML files and associated Altmetric details page are retained. This filtering process ensures that the analysis is conducted on a standardised and reliable multimodal dataset, minimising noise by incomplete crawling results and maximising the robustness of later comparative and predictive analyses. Since the focus of this project is the study of fake scientific information through retracted papers, the crawling and analysis pipeline is restricted

exclusively to retracted papers. This scope ensures that the dataset is aligned with the research objective and that all retained samples are directly relevant to the investigation of unreliable scientific content.



4.4 Deployment

This project is research-oriented and focuses on data analysis and methodological development. Rather than the deployment of a production system. Therefore, a full deployment process on client infrastructure is not required within the scope of this study. The outcomes of the project are intended to contribute to academic research and to inform the development of future systems for scientific credibility assessment. However, the methodologies and models developed in this project are designed to be reproducible and extensible. The dataset construction pipeline, feature extraction procedures, and model architectures can be adapted for integration into real-world applications, such as academic search engines, research evaluation tools, or automated content filtering systems. Future work may involve deploying these components in practical environments and implementing mechanisms for updates, maintenance, and scalability.

4.5 Testing

A detailed testing process is also not required at this stage. The current work mainly involves data collection and preparation rather than the development of a complete software system. Therefore, formal testing methods such as unit testing or integration testing are not yet central to the project. For now, the group will focus on checking the quality and consistency of the collected data.

5. RESOURCES

5.1 Hardware & Software

Several key software tools and technologies are used throughout the project. Google Drive is used as the primary cloud storage solution for managing and organizing large volumes of collected data, including MHTML files and processed datasets. It is selected due to its reliability, accessibility, and ease of integration with data processing workflows, enabling efficient sharing and backup of research data.

The Altmetric API is used to retrieve detail page data, including news headlines and related content associated with scientific papers. This API provides structured and programmatically accessible data, making it suitable for large-scale data collection and integration into the dataset. Its use allows the project to incorporate external media signals as part of the multimodal analysis.

Web crawling tools and scripts, implemented using programming languages such as Python, are used to collect MHTML webpages based on DOI and PubMed identifiers. Python is chosen due to its extensive ecosystem of libraries for web scraping, data processing, and scientific computing, which supports efficient implementation of the data pipeline.

5.2 Materials

Communication and cooperation among the group members, as a team, we have the same goal, which is completing the project. Therefore, before the project is completed, we all must actively cooperate and communicate with each other.

Communication and feedback between us and the client, as a project, we should meet the expectations of our client. Therefore, the client needs to know the progress of each stage in a timely manner. Based on the degree of satisfaction, improvement suggestions and other feedback from client, which is helpful for us to make adjustments in time.

Output standard document, if there are no time constraints, continuous communication with the clients to improve the product until the final deliverable is achieved would waste lots of time and energy. Therefore, it is necessary for us to get an output standard document, which is convenient for us to make adjustments and deliver the results.

Sufficient storage space, during the project, we will deploy high-performance computing devices with stable network connections to cope with the high concurrent requests and data writing requirements in large-scale web crawling, it is estimated that processing about 70,000 records will require tens of gigabytes of storage for web snapshots and raw JSON response data.

API access, in this project, in addition to publicly downloaded RetractionWatch datasets, projects must apply for valid Altmetric API keys (API keys and Secret) and implement strict rate limiting policies in the code, which can be used for data access, such as setting the request interval sleep time to comply with the API terms of service and prevent IP from being blocked.

The software environment, which is built entirely on the Python programming language, the core dependency libraries include Playwright for asynchronous web page crawling, Pandas for data cleansing and analysis, Requests for HTTP request handling, and Hashlib for cryptographic signatures, all dependency packages will be version-locked through the requirements file to avoid environmental conflicts.

Github, the project will use a Git version control system to manage the code base and documentation to ensure collaboration and traceability of changes between team members. All intermediate data and final results will be backed up on a regular basis, in case of data loss.

5.3 Roles & Responsibilities

For the whole semester:

- Ruicheng Zhang – team leader, coordinator and communicator, as the team leader, I am responsible for the overall coordination of the project, including task allocation, progress tracking, and making sure the group work is organised and completed on time; as a coordinator, I support collaboration within the group and with other groups when needed; as a communicator, I am responsible for maintaining communication with the client and helping ensure that the project stays aligned with the client’s expectations. I also participate in the technical aspects of the project when required.
- Tianze Xu – Librarian, records every group and client meeting into meeting minutes, and maintains proper documentation of reports.
- Thanh Le – collaborates with Gary to review all documents and reports, also takes responsibilities of making web-based UI and back-end, working with data processing and pipeline building.
- Frank Shi – Crawler Developer and Data Support, supporting data collection and preprocessing for the whole project, including the webpage crawling flow, processing identifiers, generating target URLs and improving the stability of data collection.
- Yaning Chen – the head of literature review, research analyst and data support staff, is responsible for carrying out a one-semester literature review, identifying research gaps, and assisting in building the theoretical foundation of the project. I also assist other team members to understand, sort out and verify the collected data, especially by consulting relevant records, checking texts, images and credibility-related characteristics, and assisting in interpreting data patterns for subsequent analysis.
- Haobo Zhao – completer and teamworker, as a completer, I will responsible for Pipeline Integration and System Execution, which

focus on integrating the MHTML crawling module with the Altmetric data retrieval module to ensure a unified workflow; as a teamworker, I will participate group meeting and client meeting actively, listen to other's speeches, complete the weekly tasks actively, which is assigned to me by team and write personal weekly report in time, etc.

- Han Li – responsible for the technical implementation and data collection of the project. Developed web crawling scripts and integrated the Altmetric API to gather data, while also managing dataset storage and organization using Google Drive. Also contributed to explaining the tools and methods used in the project.
- Yuqing Yang - responsible for retrieving social attention data for each retracted paper through the Altmetric API, including the Altmetric score, tweet count, news mentions, and policy citations. I cleaned and processed the data and delivered it as a structured CSV file for downstream analysis.

6. EXPECTED OUTCOMES

6.1 Project Deliverables

The client provided a dataset containing around 65,000 retracted articles with their associated metadata. Given the dataset, we would have the following expected deliverables:

- A dataset containing full detail page data of the related article retrieved from Altmetric, and it could link with the identifier in the provided dataset.
- A curated dataset containing both real and fake news articles together with their corresponding images. It would be the foundation of the project and support the analysis of textual and visual differences in the later stage.
- Textual analysis findings. These findings should be able to show how real and fake news differ in terms of linguistic patterns, sentiment, and writing style.
- Image analysis findings. We should present the techniques we used to differentiate the images used in fake news in terms of watermark usage, image resolution or quality.

- A comparative summary of the textual and visual findings across different categories of news.
- An integrated methodology to show both textual and visual perspectives together to provide a comprehensive understanding of the differences between real and fake news.
- A final report and visualisations to clearly present the findings of the project and summarise the process and outcomes achieved.

6.2 Implications

Completing this project would have beneficial implications for both the client and the general public. For the client, the findings would provide a comprehensive understanding of how to distinguish the textual and visual differences of fake news from true news. It could provide insight for the decision making on their future related research or data analysis, and might support the development of more effective frameworks or tools. Moreover, for the general public, the findings might improve their awareness of misleading information and encourage more critical reading of the online content.

7. MILESTONES / SCHEDULE

Milestone	Tasks	Reporting	Date
Week-2	Review project specification, clarify project aims	None	10-03-2026
Week-3	Identify data sources and examine the client-provided dataset	Client meeting to discuss the project aims and answer project-related questions	17-03-2026
Week-4	Develop the code for fetching Altmetric full page data	None	24-03-2026
Week-5	Proposal Report & Status Checking Due	Client meeting to review the current progress	29-03-2026
Week-6	Fetching Altmetric full detail page data to create the base dataset & perform data cleaning and preprocessing	None	07-04-2026

Week-7	Conduct an exploratory analysis of the dataset, and begin textual analysis	Client meeting to review the findings of the textual differences	14-04-2026
Week-8	Continue textual analysis and begin image analysis to examine visual differences	None	21-04-2026
Week-9	Continue image analysis and compare results across selected news categories & Progress Report Due	None	26-04-2026
Week-10	Integrate textual and visual findings to form a comparative understanding of true and fake news	None	05-05-2026
Week-11	Develop the final integrated framework and prepare visualisations, organise results	Client meeting to review the integrated framework of textual and image analysis	12-05-2026
Week-12	Documentation for final report	None	24-05-2026
Week-13	Final Report (thesis)	None	24-05-2026
Week-15	Final Presentation	None	01-06-2026

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ACKNOWLEDGEMENT

We acknowledge the use of ChatGPT (<https://chatgpt.com/>), using a combination of GPT 5.3 and 5.4, to correct the grammar mistakes. We provided ChatGPT with our draft paragraphs and asked it to refine them to be more academic.

We also used Quillbot (<https://quillbot.com/>) to paraphrase and improve the readability of certain sections.

Acknowledgement of AI in Assessment

Acknowledgement and Explanation of AI Use
Part A: Have you used AI tools in the completion of this assignment? If your answer to this part is “No”, you can leave the following Part B and Part C as blank.
Yes
Part B: What automated writing or generative AI tools you have used in the completion of this assignment? Clearly state the name(s) of the tool(s) and including a link to each tool.
ChatGPT https://chatgpt.com/
Part C: How have you used automated writing or generative AI tools in the assessment?
ChatGPT helped us plan, write, and edit our project. We largely used the software to make the structure of the research clearer, improve the language of the academic portion, and make our writing style more consistent and easy to read. It helped us adjust the Introduction, Project Problems, and related proposal elements to fit the client's new demands and the project's new scope. We selected it because it helped us edit, rewrite, and make things clearer while we were writing. When the project's direction changed close to the proposal deadline, it helped us organise our thoughts. The tool allowed us to change the words we used, move paragraphs around, and make the text more official as we were writing. Before sending in the AI output, it was looked over. The AI-generated content was included only in revised and edited form. We checked the output against the client brief, the proposal document, and the actual project context before deciding whether to use it. We then rewrote, shortened, expanded, or corrected parts of the generated text so that it matched the assignment requirements and accurately represented the project. In this way, the final submission reflects our own evaluation and adaptation of the AI-assisted content rather than uncritical copying.